Social and Ecological Sustainability of Land Improvement and Soil Use in the Japanese Agricultural Sector: Focusing on Gender Equality and Biodiversity

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Abstract The productivity of the soil in Japanese farmlands are managed and maintained by community-based Land Improvement Districts through 'Land Improvement Projects' scheme; however, their sustainability is in question. Land Improvement Districts face structural change in traditional farming practices, communities, and households. Land Improvement Projects may cause impairments to the biodiversity and rural ecology in pursuing productivity through modern rural engineering. This chapter provides insight to these issues and suggests countermeasures to secure sustainability. In particular, the chapter discusses facilitating gender equality in the Land Improvement Districts and ecologically sound practices for the Land Improvement Projects.

1 Introduction: Land (Soil) Improvement in Japan

Aizu-nōsho, one of the oldest Japanese treatises on agriculture, starts with the character of the soils 土.[1] Needless to say, soil is the fundamental of agriculture, and the Japanese word for 'land', in which the crops are sown, is 土地—consisted of kanji (Chinese characters) of 土(soil) and 地(earth).

Japan is a mountainous country, which only about 12% of the territory is arable; however, it has developed highly productive agriculture based on paddy fields. The paddy fields provide self-sufficiency in rice, which is the staple food of the Japanese

[1] *See e.g.*, Wakabayashi (2020).

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people. Extensive supplementation of fertile soil and irrigation has been required to achieve a high level of productivity of the land. For example, in one region, over 3 million cubic meters of fertile soil were scattered over the 13,000 hectares of fields in two and a half decades.[2] To provide appropriate moisture to the soil, which is also crucial to such rice cultivation, an extensive system of irrigation has also been developed throughout Japan's rural communities.

Projects for the improvement and sustainable management of the agricultural land, such as the abovementioned soil dressings and irrigation, are executed through the Land Improvement Projects 土地改良事業 ('Tochi Kairyō Jigyō') (LIPs) under the Land Improvement Act 1949.[3] Land improvement has a long history and the contemporary land improvement was shaped by the enactment of the Land Improvement Act to raise productivity of the farmlands to provide food to the people and to support the farming communities to stabilize the post-WWII land reforms.[4] The Act established LIPs as a statutory scheme, and Land Improvement Districts土地改良区 ('Tochi Kairyō Ku') (LIDs), which are associations established with the approval of prefectural governors for the purpose of implementing LIPs within the designated areas. The LIDs are the main entities to implement the LIPs and are organizations that are responsible for the management of various facilities (irrigation facilities, etc.) established by the LIPs.[5] Currently, LIDs acknowledge and publicize themselves as the keepers of water 水, soil (or land) 土, and the village里, thus calling themselves 水土里ネット. The kanji characters combined are pronounced 'midori' net[work], a similar pronunciation to the word 'green' in Japanese.[6] Despite LIDs' elegant rhyming, Japanese agriculture faces challenges in sustainability, both in the social and ecological context. Socially, Japan's agricultural industry and society are declining and aging, leading to the impairment of productivity and desolation of the communities. Environmentally, improvement of the Japanese farmlands has generally raised productivity but often at the sacrifice of biodiversity.

Along with Farmers Cooperatives, LIDs are organized by farmers in their drainage systems in conjunction with respective administrative areas to autonomously manage community's agricultural resources. Whilst the Farmers' Cooperatives work to ensure sustainability of each farm as a business entity, the LIDs are responsible for the sustainability of the basis of agriculture, including soil and water. The LID and LIP concepts have been introduced to international readers on several occasions. In 1985, volume 7 of the *Journal of Irrigation Engineering and Rural*

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[2] Hirose (2014), pp. 646–647. Such projects are called 客土('kyakudo'), or soil dressing.

[3] 土地改良法 Act No. 195 of 1949 (hereinafter 'Act').

[4] Tamaki (1977), pp. 245–246.

[5] In a large-scale LIP, the national government (Ministry of Agriculture, Forestry and Fisheries, hereinafter "MAFF") or the prefecture may become project entities instead of LIDs; however, it has to be initiated by application from farmers—usually through LIDs—except for cases provided by law, such as disaster recovery (Act art. 85, art. 87-2).

[6] Zenkoku Tochi Kairyō Jigyō Dantai Rengōkai [National Federation of Land Improvement Associations] <https://www.inakajin.or.jp/> (accessed 2025 Oct 1) (in Japanese).

Planning published a special issue on LIDs.[7] Recently, Satoh and Ishii (2021) made an excellent review on the challenges which Japanese irrigation and soil management faces. This chapter will examine LIDs and LIPs as crucial elements in the management of Japanese agriculture and soils, and in particular will focus on the involvement of women in the LIDs and the ecological and biodiversity issues in LIPs—both crucial to the sustainability of the LID and LIP scheme.

2 Organizational Characteristics of the LIDs

2.1 Ie *and* Mura

Agricultural production is established by the input of production factors such as land, labour, fixed capital, and current goods (fertilizer, etc.). The efficiency of agricultural management is enhanced through the appropriate use of these factors. Among the factors of production, agricultural lands (soil), which are rare goods that are difficult to be replicated and are often not sustainably managed by humans, are very important to farming. In particular farmland, which is of course immovable, has a long history of investment and management, and in Japan it has a presence that defines the lifestyle of farmers more than just as a factor of production. One of the characteristics of Japan's farmland is its strong community-based collectivism in use and management, which also determines the efficiency of individual agricultural management. This collectivism takes the form of *ie* いえ or 家(household) and *mura*むら or 村(village).[8] The basic unit of agricultural management in Japan, which is centred on labour-intensive rice cultivation, is not an individual but a household consisting of two~three generations of direct descendants—the *ie*. A *mura* is formed by gathering dozens of *ie* located in geographically connected areas. *Ie* is also a unit of succession of farmland (paddy fields) as a long-term improvement investment, and *mura* is a unit of mutual assistance and regulation of production and living aspects (e.g., joint management of irrigation canals that extend to the entire *mura*) that cannot be managed within the individual *ie*.[9]

[7] The volume contains 14 papers including, Nagasaki (1985), which gives a brief history of the LID and its predecessor before the enactment of the Land Improvement Act.

[8] McKean (1982) and Murakami (1985) are classical narratives of *mura* and *ie* in relation of resources management.

[9] For characteristics of *ie* and *mura*, *see* Tashiro (2003), pp. 199–205.

2.2 The LID as a Modern Legal Person

Changing the plots and characteristics of farmland and increasing productivity require targeting not only one agricultural management entity but also multiple entities working together to target farmland in a cohesive manner. The development of water conservancy facilities, which are indispensable for rice cultivation, also greatly affects the conditions for agriculture, including the availability and quality of soils, and thus the productivity of farmland. Since this water conservancy order is formed at an area level across multiple farmlands, developing related facilities in the same way as farmland is inevitable. In Japan, improving agricultural productivity requires collective response on a mura basis.

As noted above, the project to increase productivity is the LIP, based on the Act. As long as the LIP is carried out on the premise of the collectivism of farmland, the LID in charge of the project has the characteristic of an organization that incorporates a *mura* group. Since the modern *mura* itself is legally an unincorporated voluntary organization,[10] it cannot be a direct business entity. The LID scheme enables a *mura* to have legal personality to be a business entity in order to execute the LIPs.[11] The members of the LIDs, in legal terms, are clearly individual farmers.[12] In reality, however, farmers join the LIDs not as individuals, but as an *ie*, and this representative is usually the head of the household. Other members of the household are rarely members of the LIDs, even if they engage in agricultural work. In a closer look at the law, the member of the LIDs is a person who owns or has a right to cultivate farmland;[13] thus, in most cases in Japan, such rights of the household are held in the patriarch's name.

2.3 LIDs as 'Mura-Mura Associations'

The LIDs are often not simply the incorporation of individuals in a *mura*, but of a federation of several *mura*, thus referred as *mura-mura* association 村々組合 ('mura-mura kumiai').[14] The LIDs are often responsible for managing main waterways before they divert into branch waterways. From this main channel, multiple branch waterways divert, and they are further divided into terminal waterways that

[10] *Mura* discussed in this chapter are sometimes referred as 'natural *mura*', because the same word 村(Village) is used as a title for a small-scale municipality, which are sometimes called 'administrative *mura*'and almost always containing several 'natural *mura*' within its jurisdiction. See Local Autonomy Act (Act No. 67 of 1947) art. 1-3, para. 2.

[11] Issues on legal personality and memberships of the LIDs are thoroughly discussed on Kameoka (2022), pp. 345–347, which relies on Tamaki (1982), pp. 141–158.

[12] Act art. 11.

[13] Act art. 11, art. 3.

[14] *See* Tamaki (1977), pp. 239–244 for details. The word村々組合 ('mura-mura kumiai') is from Tamaki (1982), p. 149.

reach the field level one by one. The main management of the branch line and the terminal level is a water user's union 水利組合 ('suiri kumiai'), which is responsible for the maintenance and management of smaller irrigation facilities. Water user's unions are functional groups that undertake only the part related to agricultural water conservancy among the various functions of the *mura*. In some cases, this union has almost one-to-one equivalence with a *mura*, and in other cases, it is composed of multiple *mura*. Based on the characteristics of an irrigation system, which branches into small canals and ditches but does not lose its unity as a single water system, a LID normally comprehends multiple water user's unions within it.[15]

Since the fundamental purpose of the LIPs and LIDs is to improve farmlands for production, it may have negative consequences on soil fertility and biodiversity if the goals are set in short terms. *ie* and *mura*, however, are long-term institutions, which conceives to pass down fertile soil and farmland to the future generations. Thus, it should be noted that LIDs (and *ie* and *mura*) are caught between agricultural modernization policies and conservation of hereditary land, soil, and traditional practices.

3 Manifestation of *Ie* and *Mura* in LIPs: An Example of Farm Replotting

An important issue is how *ie* and *mura* exert their presence in the LIDs. In this context, it would be beneficial to consider the example of farm replotting, which is one of the types of activities under LIPs. This involves a rezoning project to change the plots and characteristics of farmland that are difficult to use, such as irregular shapes, differences in soil composition, or other unique characteristics. In conjunction with this, land ownership also changes. However, when executed through a Farm Replotting Disposition 換地処分 ('Kanchi Shobun'), which is a special legal procedure (administrative disposition), the owner of the previous farmland will become the owner of the newly configured farmland without going through complicated procedures, such as the division and consolidation of land and the transfer of ownership. Through farm replotting, productivity can be improved, and farmland accumulates smoothly. However, the selection of a replotting site may involve complex conflicts of interest and require consensus-building in implementation. In particular, farmers often have a preference to inherit the previous land even after the farm replotting, reflecting their awareness towards the farmland as household property, and this raises an intrinsic trade-off with the purpose of the project.[16] The element of *ie* greatly determines whether or not farm replotting will succeed.

[15] For case study of the multiple layers of actors and the organizational structure of the LID, see Tanaka and Sato (2003), pp. 87–88.

[16] *See* Nakajima and Arimoto (2011).

In addition, the physical transformation of farmland forces a change in the practices and customs that have been formed on the farmland. Many customary rules and practices have accumulated through the long history of collective use of land and facilities for agriculture. Land improvement investments in these may force a change in local practice. Since customs are the very identity of local communities, investments are not always carried out as a matter of course, even if they are economically rational.[17] In other words, farm replotting is a project that is basically undertaken on a *mura* basis as it targets the land of the *mura*, and the project also modifies the community order and customs of the *mura*.

Farmland consolidation which causes land replotting contributes to preventing soil erosion and improving water retention in farmland.[18] On the other hand, farmland consolidation which aims for efficient farming through modern agriculture, may have the adverse effects to biological sustainability, which will be discussed in Sect. 5.

4 Can the Appointment of Female Directors Strengthen the Organizational Foundation of the LIDs?

4.1 Transformation of ie *and* mura

As discussed earlier, *ie* and *mura* are the foundation of the LIDs and important regulatory factors of the LIPs. At present, however, the transformation of *ie* and *mura* is inevitable. Naturally, that transformation will have a significant impact on the LIDs.[19] First of all, it is clear that the symptoms indicating the crisis of Japan agriculture—aging of farmers, decline of farm population, farm retirement, increase of land tenure non-farm households, and occurrence of abandoned farmland—are all connected to the ebb of *ie*. Because of the decrease of farming income, agriculture has lost its status as a successful family business. Agriculture has long been borne only by a specific individual amongst family members. Long-standing agricultural structural policies have created many of the agricultural holdings to expand to a management scale of several tens of hectares. These entities often incorporate and use employed labour; hence they can no longer be classified as *ie* farming. In addition, the geographical area of large-scale holdings (often cooperate) often exceeds the scope of a single *mura* and encompass multiple *mura*. This means that a large-scale holding may be involved in the land use and irrigation system of several *mura* that that holding does not originally belong.

[17] Nakajima (1998), p. 197.

[18] Miyagi Ken Tochi Kairyō Jigyō Dantai Rengōkai [Miyagi Prefecture Federation of Land Improvement Association]<https://mlw.or.jp/business/improvement/> (accessed 2025 Oct 1) (in Japanese).

[19] Satoh and Ishii (2021), pp. 429–431.

Thus, the self-contained *mura*, which consisted of *ie* as homogeneous family-owned self-employed family farms, has been transformed into a space that allows for trans-boundary activities by heterogeneous entities. As a result of this diversification of interests of the members, consensus building has become difficult. In terms of the LIPs, it has also become increasingly difficult to procure the consent of more than two-thirds as required by law.[20] Amongst the current *mura*, there are self-employed farmers, leased land farmers who have expanded on a correspondingly large scale, and land tenure non-farm households. In such a situation where a *mura* is becoming more heterogeneous and the level of land rent is not expected to rise, smooth decision-making regarding land improvement investment, maintenance, and management of facilities is difficult.[21] Conditions for the functioning of a *mura* as a spontaneous but integrated and coordinated system are therefore being undermined.[22]

The most important impact on the LIDs by the changes in *ie* and *mura* described above is the deterioration of land improvement facilities. If the members of the LIDs retire from farming, even if they continue to belong in the LIDs, they will lose the incentive to proactively develop land improvement measures and infrastructure. The same is true for the collective facility management at a *mura* level. Hence, the activities of the LIDs will stagnate. As a result, infrastructure and facilities are aging, and the continuation of farming is hindered, especially in mountainous areas. This trend is expected to gradually extend to the plains. In a partial amendment to the Land Improvement Act in 2025, a provision was added to enable the MAFF or the prefecture to conduct LIPs for facility modification without farmers' application,[23] which is a prerequisite before.[24] However, increasing productivity through the LIPs is a basic prerequisite for the economic viability of agricultural production activities. If the goal of agricultural policy is to supply "a stable supply of food to the people … based on increase of domestic agricultural production",[25] then the significance of the LIPs is greatly increased. Needless to say, the survival of the LIDs is an important matter impacting the sustainability of agricultural production in Japan.

[20] This has already been pointed out by Imamura (1977), p. 365.

[21] Onimaru (2018), pp. 524–526.

[22] Satoh and Ishii (2021), p. 431.

[23] Art. 87-2, para. 1, item 3 of the revised Act, the amendment was enacted on 31 Mar 2025 and enforced from 1 Apr 2025.

[24] See *supra* note 5.

[25] Basic Act on Food, Agriculture and Rural Areas (Act No. 106 of 1999) art. 2, para. 2. Japan Law Translation <https://www.japaneselawtranslation.go.jp/ja/laws/view/4984> (accessed 1 Oct 2025).

4.2 Promoting the Appointment of Female LID Directors

The current LIDs are affected by the decline in the number of farmers, and they are often struggling to elect directors. How then can the organizational base of the LIDs be strengthened?

Currently, many LIDs are considering the appointment of female directors. It has long been said that Japan is lagging behind in women's participation in society, the agricultural sector has been particularly slow to implement such participation. Again, this is because the basic unit of agricultural production in Japan has traditionally been *ie*, and it has been common for men who are the heads of households to perform the external roles. It is usually the man who holds the property rights to the agricultural land of the *ie*; whether it is ownership or the right to cultivate. The majority of the members of the current LIDs is men. Despite the fact that women are engaged in actual agricultural work as members of *ie*, women are usually not members of the LIDs at the outset. Women may become the members of the LIDs in secondary circumstances, such as when women inherit the ownership of farmlands from the deceased husband. The LID directors may be required to engage in physical labour (e.g., operation of sluice gates during typhoon), and these have been considered difficult for women.[26]

The current policy platform, the Fifth Basic Plan for Gender Equality (2020),[27] sets a target of increasing the percentage of female directors in the LIDs nationwide from only 0.6% in fiscal 2016 to 10% in fiscal 2025. At the time of writing, this goal will not be achieved, but as of April 1, 2025, it has risen to 2.7% (1,174 out of 42,953).[28] Although the percentage of female directors in the LIDs was much lower than that of agricultural cooperatives (JA), which is aiming to be more than 10% in the same year, the number of female directors is gradually increasing. In cases where it is difficult to elect a woman as a director from within a LID due to the absence of formal female members, the non-member director system is used.[29] Candidates for non-member positions include the families of members and women who serve on the boards of agricultural cooperatives. Of the total 1,174 female directors nationwide as of April 1, 2025, 621 are internal directors and 553 are non-member directors.[30]

[26] Maki (2024), p. 7.

[27] Gender Equality Bureau (2023), p. 49.

[28] Zenkoku Tochi Kairyō Jigyō Dantai Rengōkai, Gender equality in land improvement organizations <https://www.inakajin.or.jp/gender-equality#anc01> (accessed 2025 Oct 1) (in Japanese).

[29] According to the Land Improvement Act, at least three-fifths of the directors of a LID must be members; accordingly, the remaining two-fifths can be non-members (Art. 18, para. 5). This may require revising provisions in the charter of the LIDs.

[30] Zenkoku Tochi Kairyō Jigyō Dantai Rengōkai, at *supra* note 28.

4.3 Effects of the Appointment of Female Directors to the Board: Restructuring of ie *or Transformation into a New Association*

One of the most common direct purposes or reasons cited for appointing female directors is to solve the shortage of board members. In a situation where a shortage of directors threatens the formal existence and survival of the LIDs themselves, the appointment of female directors is a means to compensate for this. Some may view the appointment of female directors as a liberal trend that will significantly change the male-dominated *ie* and *mura*. However, it may equally be viewed as a conservative trend to maintain the existing framework of *ie* and *mura* with the help of women. Trying to protect the old *ie* and *mura* system may be in subtle contrast with the efforts of large-scale agricultural holdings: growing beyond the framework of *ie* and *mura* and becoming prominent in Japanese agriculture.[31] In extreme cases, female directors may be appointed simply for the purpose of matching the number of board members. However, the appointment of female directors is said to have the effect of better utilising human resources, improving the quality of services for association members, and improving the reputation of local residents.[32] In addition, it may be said that a diverse network of female directors may have a positive impact on the LIDs.[33]

It has long been the view that as long as traditional *ie* and *mura* are keep intact, the LIDs do not need to have contact with the external environment. However, that is not the case today. The modern LIDs are gaining access to the external environment and creating the conditions for facilitating adaptative actions through their female directors. As an example, social equity has become important for the LIDs to gain support from the general public, which was not the case in the past. LIDs can also acquire means to actively influence the external environment through female directors, who have channels hitherto unknown to the male-oriented LIDs.[34] The increase in the number of female directors can bring about changes not only in the LIDs itself but also in the rural society at large surrounding the LIDs.

Traditionally, the LIDs have had a strong foundation of *ie* and *mura*, but that premise has changed and the LIDs are facing an existential crisis. If we want to sustain agricultural production in Japan, we cannot underestimate the LIDs, which have played an essential role in improving agricultural productivity. At present, the sustainability of the LIDs is strongly questioned: the LIDs must respond in a realistic way to the transformation of *ie* and *mura*. In 2025, amendment of the Land Improvement Act, a provision was added to require LIDs 'to give consideration to

[31] Satoh and Ishii (2021), p. 432.

[32] Ministry of Agriculture, Forestry and Fisheries (2023), p. 10.

[33] *See* Maki (2024), p. 10.

[34] Similar arguments are raised on fishery cooperatives by Onaka (2023), p. 524.

avoid unbalanced age and sex representation on the boards.'[35] Although there are still very few cases nationally, we must pay attention to the appointment of female directors and observe how that can contribute to the sustainability of not only the association but the rural society at large, and ultimately contribute to improve conservation of farmland and soil.

5 Rural Ecology and Land Improvement

Although LIPs as well as the LID system are designed for production, they have contributed to sustaining the paddy fields, which is not only crucial to the ecology of the Japanese archipelago but provides multiple services to the society. In 2001, the Science Council of Japan estimated the annual services provided by farmland in Japan amounts to 5.8 trillion yen, including 3318 million yen in prevention of soil erosion.[36] Furthermore, special LIPs are executed to rid the soil of agricultural land contamination from designated hazardous substances. One prominent example are the LIPs executed in the Toyama Prefecture covering over 1500 ha of paddy fields contaminated by cadmium, which was caused by mass poisoning known as Itai-itai Disease.[37] The Land Improvement Act was revised in 2001 to require consideration to the environment in executing LIPs (Act art. 1, para. 2). Recently, the government is promoting the positive environmental effects of the paddy fields, such as providing amenity through waterways and scenery, flood prevention, soil protection, absorbing excessive warming, conserving water, and providing habitat for biodiversity.

Taking a closer look, however, LIPs have caused impairments to the ecology of rural Japan. Their negative effects on biodiversity are multi-faceted.[38] LIPs have modified conventional small-scale paddy fields, where thick mud was covered by shallow water year-round, to larger scale paddy fields where water is provided only when the crop requires. Whilst contributing to reducing labour costs and raising production, such consolidation and drainage improvement turns the soil as well as the irrigation ditches dry for most of the year, reducing aquatic species and degrading the ecosystem. Not only fish life, which has a long list of endangered species due to the lack of water by modernization of the irrigation systems, but also species such as Chinese mystery snails (*Cipangopaludina chinensis laeta*)—a popular food source in the past—which depend on muddy soil for survival, have become endangered due to the aridification of the soil by irrigation improvements.[39] Further

[35] Act art. 18, para. 6 of the revised Act.

[36] Science Council of Japan (2001). The estimate is still limited as it does not include services such as providing wildlife habitat, nor discharging nutrient salt to coastal areas, both of which can have negative effects as well as positive ones.

[37] *See e.g.*, Kawai (2017).

[38] *See generally*, Natuhara (2013).

[39] Nakai (2015), p. 275.

refinements of LIPs, especially minimizing the adverse effects to biodiversity and soil sustainability, are necessary to maximize their contribution to Japan's rural ecology. Reacting to such calls, LIDs are increasingly engaging in actions to facilitate habitats for wildlife and to take adopt more ecologically friendly tactics. Notable examples can be found in Toyooka in western Japan where a reintroduction project for endangered Oriental White Storks (*Ciconia boyciana*—an endangered species protected by statutes) is being developed. The irrigation waterways which were inaccessible for fishes after the LIP have been reconstructed to allow fish passage and year-long flooding of paddies has been re-introduced.[40] Winter-flooding of the paddies not only benefits wintering waterfowls, droppings of birds enrich the soil.[41] Irrigating during non-farming season; however, poses a challenge for LIDs as their water rights is limited to agricultural purposes, and it will be against customary practice.

Such projects may still be biased towards conservation of charismatic endangered species,[42] and the depth of involvement of the LIDs in ecological projects varies, with many LIDs not having any involvement.[43] Nevertheless, such environmental initiatives by LIPs and LIDs need to expand for the sustainability, not only of biodiversity but for the sustainability of the LID itself by gaining support in the community.[44]

6 Conclusion

Sustainability of the LIDs, which is an important factor in Japanese agriculture through executing LIPs and maintaining their achievements, is under challenge. The main challenge comes from socio-economic factors impacting on the agricultural sector. However, the ecological environment is also of serious concern. Negative impacts to ecology resulting from LIPs are likely to turn the public against LIDs and deter public support, remembering that LIDs are subsidized heavily from tax moneys.

One of many strategies those of which are responsible for the promotion and implementation of LIDs are following to meet these challenges is to promote female board members. Promoting female involvement may change not only the composition of a LID but also the rural community itself. Another development is to introduce ecologically sound practices, which may change the on-going conduct and management of agriculture in Japan to strengthen the sustainability of the farming industry and public support. Incorporating these strategies will hopefully help to sustain and enhance farmland with its healthy soil for the future generations.

[40] Naito and Ikeda (2007), p. 220; Ootagaki (2010), pp. 490–491.

[41] Yamamoto et al. (2003).

[42] Mori (2011), p. 168.

[43] Yoshinaga et al. (2018), p. 1013.

[44] *Id*. p. 1014.

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